Growth Accounting Cheatsheet

Macroeconomics I - Lecture 3

January 2026

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1 Notation and Definitions

Core Variables

- Y_t: Aggregate output
- K_t: Aggregate capital stock
- L_t: Labor force
- B_t: Technology level (TFP)
- A_t: Labor productivity (often used interchangeably with B_t)
- H_t: Human capital-adjusted labor ($H_t = e^{\phi(E_t)} L_t$)
- E_t: Years of schooling
- w_t: Real wage
- r_t: Real interest rate
- δ: Depreciation rate
- I_t: Investment

Growth Rates

- $\gamma_Y \equiv \dot{Y}/Y$: Output growth rate
- $\gamma_A \equiv \dot{A}/A$: TFP growth rate
- $\gamma_K \equiv \dot{K}/K$: Capital growth rate
- $\gamma_L \equiv \dot{L}/L$: Labor growth rate

Key Ratios

- $\frac{K}{Y}$: Capital-output ratio
- α: Capital share (typically 1/3)
- $1 - \alpha$: Labor share

2 Growth Accounting Framework

Primal Approach: Basic Setup

Production function (CRS):

$$Y_t = F(K_t, B_t L_t)$$

Taking logs and time derivatives:

From production function to growth accounting:

$$\frac{\dot{Y}}{Y} = \frac{F_K K}{F}\frac{\dot{K}}{K} + \frac{F_L L}{F}\frac{\dot{L}}{L} + \frac{F_B B}{F}\frac{\dot{B}}{B}$$

Define output elasticities:

$$\alpha_t = \frac{F_K K}{F}, \quad 1 - \alpha_t = \frac{F_L L}{F}$$

Under CRS (Euler's theorem): $\alpha_t + (1 - \alpha_t) = 1$

$$\boxed{\gamma_Y = \gamma_A + \alpha_t \gamma_K + (1 - \alpha_t)\gamma_L} \qquad \text{(Growth-Accounting)}$$

Solow Residual

Everything except measured capital and labor goes into TFP:

$$\boxed{\gamma_A = \gamma_Y - \alpha_t \gamma_K - (1 - \alpha_t)\gamma_L} \qquad \text{(Solow-Residual)}$$

Components of γ_A (what's inside the black box):

1. True technology progress
2. Misallocation (market efficiency, regulations, cronyism)
3. Mismeasurement of capital and labor quality
4. Returns to scale violations
5. Externalities

3 Measuring Capital Stock

Perpetual Inventory Method

Law of motion for capital:

$$K_{t+1} = (1-\delta)K_t + I_t \tag{PIM-Motion}$$

Where:

- I_t: Investment (from national accounts)
- δ: Depreciation rate (easy to estimate)
- K_0: Initial capital stock (most complicated part)

With constant growth assumption for initial period:

$$K_0 = \frac{I_0}{g+\delta} \tag{PIM-Initial}$$

where g is the steady-state growth rate.

4 Cobb-Douglas Special Case

With Cobb-Douglas production $Y = K^\alpha (BL)^{1-\alpha}$:

$$\begin{aligned} \gamma_Y &= \gamma_A + \alpha\gamma_K + (1-\alpha)\gamma_L \\ \gamma_A &= \gamma_Y - \alpha\gamma_K - (1-\alpha)\gamma_L \end{aligned} \tag{CD-Growth-Accounting}$$

Alternative decomposition (output per capita):

$$\frac{Y}{N} = B\frac{L}{N}\left(\frac{K}{Y}\right)^{\frac{\alpha}{1-\alpha}} \tag{CD-Decomposition}$$

Growth rates:

$$\gamma_{Y/N} = \gamma_B + \gamma_{L/N} + \frac{\alpha}{1-\alpha}\gamma_{K/Y} \tag{CD-PerCapita}$$

5 Empirical Results

Historical Growth Accounting (1947-1990)

Country	γ_Y	$\alpha\frac{\gamma_K}{\gamma_Y}$	$(1-\alpha)\frac{\gamma_L}{\gamma_Y}$	$\frac{\gamma_A}{\gamma_Y}$
		1947-1973		
US	4.02%	42.7%	23.7%	33.6%
UK	3.73%	47.2%	0.9%	51.9%
France	5.42%	41.5%	3.9%	54.5%
Germany	6.61%	40.6%	2.8%	56.6%
		1960-1990		
US	3.10%	45.2%	41.5%	13.2%
UK	2.49%	52.3%	-4.2%	51.9%
France	3.50%	58.1%	0.5%	41.4%
Germany	3.20%	58.7%	-8.1%	49.4%

Key finding: TFP accounts for 1/3 to 1/2 of output growth (not 70% as early studies suggested).

US Decomposition (Jones 2015)

Period	Y/N	B	L/N	K/Y
1948-2013	2.5	2.0	0.3	0.1
1948-1973	3.3	3.2	0.3	-0.2
1973-1995	1.6	0.8	0.4	0.4
1995-2007	2.8	2.2	0.3	0.2
2007-2013	1.7	1.1	0.5	0.1

Key Observations:

- Capital deepening of little importance (US near BGP)
- Large importance of Solow residual
- Productivity slowdown in 1970s-80s and post-2007

6 Dual Approach

From Primal to Dual

Starting from national accounting identity (no profits):

$$Y_t = w_t L_t + r_t K_t$$

Taking log derivatives:

$$\frac{\dot{Y}}{Y} = \alpha_K \left(\frac{\dot{r}}{r} + \frac{\dot{K}}{K}\right) + \alpha_L \left(\frac{\dot{w}}{w} + \frac{\dot{L}}{L}\right) \tag{Dual-Accounting}$$

TFP growth via dual approach:

$$\boxed{\gamma_A = \alpha_K \frac{\dot{r}}{r} + \alpha_L \frac{\dot{w}}{w}} \tag{Dual-TFP}$$

Comparison: Primal vs. Dual

- Primal: Uses production function elasticities ($F_K K/F$)
- Dual: Uses factor cost shares (rK/Y, wL/Y)
- Should coincide in theory, differ in practice
- Dual is more robust (no functional form assumptions)

Dual vs Primal: Country Evidence

Country	Primal	Dual
Hong Kong ('66-'91)	2.3%	2.7%
Singapore ('72-'90)	-0.7%	2.2%
South Korea ('66-'90)	1.7%	1.5%
Taiwan ('66-'90)	2.1%	3.7%

7 Factor Shares vs. Elasticities

The Key Insight

Under perfect competition, factor shares equal output elasticities:

$$\frac{F_x x}{Y} = \frac{w_x x}{pY} \tag{Share-Elasticity}$$

This is why we can use measured income shares in growth accounting.

Problem 1: Market Power

With monopoly power:

$$\frac{\epsilon_D - 1}{\epsilon_D} \frac{F_x x}{Y} = \frac{w_x x}{pY} \tag{Market-Power}$$

The markup is $\mu = \frac{\epsilon_D}{\epsilon_D - 1}$.
With profits, elasticity = cost share (not revenue share).

Problem 2: Fixed Costs

If there are fixed costs Γ:

$$\frac{F_x x}{Y} = \frac{w_x x}{pY} \neq \frac{w_x (x + \Gamma)}{pY} \tag{Fixed-Costs}$$

Factor share $\neq$ elasticity when fixed costs exist.

Problem 3: Externalities

Private marginal product $\neq$ social marginal product with externalities.

Required conditions for Share = Elasticity:

1. Perfect competition (no market power)
2. Variable costs only (no fixed costs)
3. No externalities (private = social MP)

8 Development Accounting

Framework

Cross-sectional comparison at a point in time:

$$\log Y_i = \log A_i + \alpha \log K_i + (1-\alpha) \log L_i \tag{Dev-Accounting}$$

Goal: Decompose income differences into:

- Factor quantities (K, L)
- TFP (A)

Key result (Caselli 2005): TFP differences account for $> 50\%$ of income variation.

Sample	obs	y_{90}/y_{10}	Variance Explained
Whole sample	105	24.8	61.5%
OECD	24	1.7	39.4%
Non-OECD	81	14.7	64.4%

9 Hall-Jones Framework

Production Function

$$Y_i = K_i^{\alpha}(A_i H_i)^{1-\alpha} \tag{HJ-Production}$$

where:

- K_i: Physical capital
- $H_i = e^{\phi(E_i)} L_i$: Human capital
- A_i: Labor productivity (TFP)
- E_i: Years of schooling

Returns to Education

$$\phi(s) = \begin{cases} 0.134 \times s & \text{first 4 years} \\ 0.134 \times 4 + 0.101 \times (s-4) & \text{years 4-8} \\ 0.134 \times 4 + 0.101 \times 4 + 0.068 \times (s-8) & \text{year 9+} \end{cases} \tag{Returns-Education}$$

Output per worker:

$$y_i = \left(\frac{K_i}{Y_i}\right)^{\alpha/(1-\alpha)} h_i A_i \tag{HJ-PerWorker}$$

Measure TFP as residual:

$$A_i = \frac{y_i}{\left(\frac{K_i}{Y_i}\right)^{\alpha/(1-\alpha)} h_i} \tag{HJ-TFP}$$

Hall-Jones Results (Ratios to US)

Country	Y/L	$(K/Y)^{\alpha/(1-\alpha)}$	H/L	A
United States	1.000	1.000	1.000	1.000
Canada	0.941	1.002	0.908	1.034
Italy	0.834	1.063	0.650	1.207
West Germany	0.818	1.118	0.802	0.912
France	0.818	1.091	0.666	1.126
UK	0.727	0.891	0.808	1.011
Hong Kong	0.608	0.741	0.735	1.115
Singapore	0.606	1.031	0.545	1.078
Japan	0.587	1.119	0.797	0.658
Mexico	0.433	0.868	0.538	0.926
India	0.086	0.709	0.454	0.267
China	0.060	0.891	0.632	0.106
Average, 127 countries	0.296	0.853	0.565	0.516

Key finding: Most income differences come from TFP (A).
Decomposition of top5/bottom5 gap:

$$y_{top5} = 1.8 \times 2.2 \times 8.3 \times y_{bottom5} \tag{Decomposition}$$

(Capital $\times$ Human Cap $\times$ TFP)

10 Institutions and TFP

Hypothesis: Better institutions $\Rightarrow$ higher TFP $\Rightarrow$ higher income.
Basic regression:

$$\log y_i = \gamma + \beta S_i + \epsilon_i \tag{Institutions-Reg}$$

Problem: Reverse causality (income $\rightarrow$ institutions).
Solution: Instrument for institutions with:

- Distance from equator
- Percentage speaking Western languages
- Predicted trade share (Frankel-Romer)

Results (Hall-Jones 1999):

Specification	Social Infrastructure	$\hat{\sigma}_z$	OverID p-value
Main specification	5.14	0.84	0.256
Instruments: Dist, FR	5.00	0.82	0.208
OLS	3.29	0.70	—

Conclusion: Institutions (social infrastructure) strongly correlated with TFP and output per worker.

11 Misallocation: Hsieh-Klenow (2009)

Setup

Monopolistic competition with heterogeneous productivity A_i:

$$y_i = A_i k_i^{\alpha} l_i^{1-\alpha} \tag{HK-Production}$$

CES preferences over varieties:

$$C = \left(\int_i c_i^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} \tag{CES-Preferences}$$

Optimal pricing (monopolistic competition):

$$p_i = \frac{\sigma}{\sigma-1} \frac{x}{A_i} \tag{Pricing}$$

where marginal cost $x = \left(\frac{r}{\alpha}\right)^{\alpha} \left(\frac{w}{1-\alpha}\right)^{1-\alpha}$.

Output relative to productivity:

$$y_i \propto A_i^{\sigma} \tag{HK-Output}$$

Distortions

Add firm-specific distortions:

$$\max_{y_i, k_i, l_i} (1-\tau_i^Y) p_i y_i - w l_i - (1+\tau_i^K) r k_i \tag{Distorted-Problem}$$

Distorted capital-labor ratio:

$$\frac{k_i}{l_i} = \frac{\alpha}{1-\alpha} \frac{w}{r} \frac{1}{1+\tau_i^K} \tag{Distorted-KL}$$

Distorted output:

$$y_i \propto A_i^{\sigma} \left(\frac{1-\tau_i^Y}{1+\tau_i^K} \right)^{\sigma} \tag{Distorted-Output}$$

TFPR and TFPQ

Define Total Factor Productivity Revenue:

$$TFPR_i \equiv p_i A_i = \frac{\sigma}{\sigma-1} x \frac{(1+\tau_i^K)^{\alpha}}{1-\tau_i^Y} \tag{TFPR}$$

Aggregate TFPQ:

$$A = \left[\int_j \left(A_j \frac{TFPR}{TFPR_j} \right)^{\sigma-1} \right]^{\frac{1}{\sigma-1}} \tag{TFPQ}$$

Key insight: Dispersion in $TFPR$ reduces aggregate productivity.

12 HK Counterfactuals

TFP gains from equalizing TFPR within industries:

Country	Early		Mid		Late	
	Year	% Gain	Year	% Gain	Year	% Gain
China	1998	115.1	2001	95.8	2005	86.6
India	1987	100.4	1991	102.1	1994	127.5
United States	1977	36.1	1987	30.7	1997	42.9

Country	Early		Mid		Late	
	Year	% Gain	Year	% Gain	Year	% Gain
China	1998	50.5	2001	37.0	2005	30.5
India	1987	40.2	1991	41.4	1994	59.2

Interpretation: Large TFP gains possible if India/China had US-level allocation of resources across firms.

13 Common Pitfalls

1. **Confusing factor shares with elasticities:** Only equal under perfect competition, no fixed costs, no externalities.
2. **Attributing causality:** Growth accounting is accounting, not causation. Capital accumulation may itself be driven by TFP.
3. **Ignoring quality changes:** Labor quality (education) and capital quality (equipment quality) matter significantly.
4. **Using wrong growth rates:** Remember technology growth x affects effective variables.
5. **Measuring capital incorrectly:** Perpetual inventory requires good initial K_0 and proper δ.
6. **Treating TFP as pure technology:** It contains misallocation, measurement error, and externalities.
7. **Assuming Cobb-Douglas:** Elasticity of substitution = 1 may not hold.
8. **Ignoring dual-primal discrepancies:** Different methods give different results in practice.

14 Quick Reference Formulas

Growth Accounting:	$\gamma_Y = \gamma_A + \alpha\gamma_K + (1-\alpha)\gamma_L$
Solow Residual:	$\gamma_A = \gamma_Y - \alpha\gamma_K - (1-\alpha)\gamma_L$
Perpetual Inventory:	$K_{t+1} = (1-\delta)K_t + I_t$
Initial Capital:	$K_0 = I_0/(g+\delta)$
CD Output/Worker:	$y = (K/Y)^{\alpha/(1-\alpha)}hA$
HK TFPR:	$TFPR_i = p_i A_i$
TFPQ:	$A = \left[\int A_j^{\sigma-1}\right]^{1/(\sigma-1)}$
Returns to Ed:	$\phi(s) = 0.134s$ (first 4 yrs)

15 Problem-Solving Templates

Template 1: Basic Growth Accounting

1. Identify production function and parameters
2. Compute growth rates of Y, K, L
3. Use factor share α (typically 1/3)
4. Calculate Solow residual: $\gamma_A = \gamma_Y - \alpha\gamma_K - (1-\alpha)\gamma_L$
5. Interpret: what fraction of growth comes from TFP vs. factors

Template 2: Development Accounting

1. Get data on Y/L, K/Y, H/L for countries
2. Use $\alpha = 1/3$
3. Compute $A_i = y_i/[(K/Y)^{\alpha/(1-\alpha)}h_i]$
4. Compare TFP across countries
5. Decompose income gap into factors vs. TFP

Template 3: Misallocation Analysis (HK)

1. Estimate TFPR dispersion across firms
2. Compute aggregate TFPQ from firm productivities
3. Calculate counterfactual: equalize TFPR
4. Compute TFP gain: $(A_{counterfactual}/A_{actual}) - 1$

16 Implementation Notes

Packages Used

- amsmath, amssymb, mathtools: Mathematics typesetting
- graphicx: Graphics support
- xcolor: Color definitions
- siunitx: Number formatting
- cleveref: Cross-references
- hyperref: Hyperlinks
- tcolorbox: Colored derivation and result boxes
- booktabs: Professional tables
- enumitem: List formatting

Custom Notation

- `\gammay`: Output growth rate γ_Y
- `\gammaA`: TFP growth rate γ_A
- `\ky`: Capital-output ratio K/Y
- `\tfpr`: Total Factor Productivity Revenue
- `\tfpQ`: Total Factor Productivity Quantity

Compilation

Compile with: `pdflatex -interaction=nonstopmode Growth_Accounting_CheatSheet.tex`
Run twice to resolve cross-references.